

Examiner reports

Q1.

This question was generally well attempted with over a quarter of students achieving full marks. Students recognised the need to expand $f'(x)$, integrate their expansion and evaluate the constant of integration from the given condition. Unfortunately the expansion proved to be too demanding algebraically for some, with incorrect squaring of the $2x$ and

$$\left(-\frac{3}{x}\right)$$

terms the major issue.

The integration was generally well done, although some students had issues with the negative power and some omitted the constant of integration which prevented them from completing the solution.

Q2.

Part (a) was well attempted and many students scored full marks. The question left students with the choice of how to approach the proof and inefficient but correct methods were often seen. Students who scored one mark often failed to complete the proof. Students should be in the habit of finishing any proof or “show that” question with a concluding statement which completes their argument.

While many students scored full marks on part (b) the most efficient method was rarely seen. Students who solved $p(x) = 0$ on their calculators and made use of the factor theorem to write down the correct factorisation could achieve full marks in one line.

Part (c) discriminated well between students of varying abilities. Many students could start the question by manipulating the given equation to form a cubic equation. Some students who went on to see the connection with the earlier part of the question made errors with their use of inequalities or failed to complete their argument with enough rigor.

Some students seemed to focus on the phrase “no real solutions” and were determined to try and use the discriminant of an incorrect quadratic.

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Q5.

- (a) (i) Although there were still some who thought that the Factor Theorem involved long division, the vast majority found $p(3)$ and showed sufficient working to verify that $p(3) = 0$. There was still a significant number who failed to include a concluding statement such as “therefore $x - 3$ is a factor” and these students did not score full marks.
- (ii) The majority used long division or inspection to find the quadratic factor and then successfully factorised to be able to express $p(x)$ as a product of linear factors. The repeated factor seemed to disturb many students who felt that the factorised form must be something like $(x - 3)(x - 2)(x + 3)$.
- (b) A generous method mark was awarded for drawing a recognizable cubic curve. Those with the wrong factors usually earned this mark. Those with the correct factors were sometimes unsure of the correct shape of the curve in the vicinity of $x =$

3. It was pleasing to see many good sketches, but these were occasionally spoiled by not marking the values -2 and 3 on the x -axis or by poor curvature beyond $x = 3$ or before $x = -2$.

Q6.

This question provided a good start for the students with about 75% of the entry gaining 5 or more of the 7 marks available. However, many did not use the most efficient algebraic methods and will have paid a time penalty later in the paper.

Part (a). Most students gave a correct answer showing clearly their use of the Remainder Theorem. Those who used long division gained no credit.

Part (b)(i). A full justification was required to gain the mark, but some students just wrote $d = 3$ and the given answer. An uncommon error was to find d as -4 .

Part (b)(ii). Students were expected to find the value of a by inspection, although some used long division or equating coefficients. Some students did not respond to the request to write $g(x)$ as a product of three linear factors until it was required in part (iii).

Part (b)(iii). Many students took the expected route of fully factorising the denominator and cancelling down the common factors with those of $g(x)$ and simplifying the result. Other methods, such as multiplying by the denominator and equating coefficients or using long division, were usually successful but unnecessarily time consuming.

Q7.

- (a)
 - (i) Almost all candidates attempted to find $p(-3)$ and most were successful in showing that $p(-3) = 0$. It was pleasing to see many candidates taking heed of previous reports and writing a concluding statement such as “so $x + 3$ is a factor”. Those who simply used long division were ignoring the request to use the Factor Theorem and earned no marks.
 - (ii) The majority of candidates used inspection or long division and found the correct quadratic factor. A few using the latter method did not always write their final answer as the product of factors as requested.
- (b)
 - (i) Almost all found the first derivative correctly, though a few added “+ c ”.
 - (ii) Several candidates presented an inadequate solution here. It was necessary to both equate their answer to part (b)(i) to zero and divided by 4 in order to obtain the given equation.
 - (iii) Solutions to this part were very disappointing. Most candidates merely substituted $x = -3$ into the given equation and were, in effect, repeating part (a). Those who attempted to solve the quadratic equation or found the value of the discriminant did not always give a satisfactory explanation for the single stationary point of the curve. For example “the discriminant is negative therefore the only real root is $x = -3$ ”.
 - (iv) Once again the second derivative was found correctly by most, although its value was sometimes incorrect as a result of poor arithmetic, particularly in the

evaluation of the term $12(-3)^2$. Unfortunately, quite a few candidates divided their expression for $\frac{dy}{dx}$ by 4 before differentiating again and wrote $\frac{d^2y}{dx^2} = 3x^2 - 4$, which was incorrect.

- (v) The interpretation of the sign of the second derivative was well known. However not all candidates gave an adequate reason. A statement such as “ $92 > 0$ ” or “ $\frac{d^2y}{dx^2}$ is positive” was required; an answer such as “it is positive” or “the value is greater than 0”, without specific reference to the value from part (iv), was not sufficient.

Q8.

This question was generally done well with about 35% of the entry gaining 10 or 11 marks. Often the mark lost was in part (a) for failure to show any processing or to give a conclusion.

- (a) Virtually all students knew they were to substitute $x = -\frac{3}{2}$, and evaluate. However, those who just substituted into $f(x)$ and wrote $= 0$ without showing any stages in the arithmetic did not score the accuracy mark, nor did those who did not interpret the $= 0$ result.
- (b) Most students had this correct and virtually all scored at least the method mark. Some students did the factorisation by inspection whilst long division was also a popular method. Those students who chose to use undetermined coefficients often did much unnecessary work, as at least the values of a and c should be apparent from inspection.
- (c) (i) This part of the question was also generally done well with most students able to quote the cosine double angle formula or derive it correctly, and convert it to sines. A rare error was a sign error or the dropping of the 2. Some students did not handle multiplying out the expression convincingly, and as the required result was given, they were penalised. However, most did convert their trigonometric expression into the required equation in x .
- (ii) Some students did not attempt to solve this equation, but of those who did they virtually all knew they had to solve the cubic equation in x . Many did this successfully, often following a failed attempt at factorisation. Some made sign or other errors in their use of the quadratic formula. Some students used the quadratic formula on the cubic equation itself. Students who had not used the quadratic formula correctly with the given equation, could gain no further marks. However, many did complete successfully and it was pleasing to see the inappropriate roots of the cubic equation carefully rejected. Some students did not correctly convert the -16.3° they obtained from their calculator into angles in the required range.

Q9.

$\frac{3}{4}$

In part (a), virtually all candidates knew that they were to substitute $x = \frac{3}{4}$ into the cubic expression, but many either failed to show any detail of a calculation leading to zero and / or failed to give a conclusion as to the significance of such a result. Candidates need to recognise that when a printed answer is given, as with $(4x - 3)$ being a factor here, they need to give a full justification of why this is the case.

In part (b), most candidates showed that they were familiar with the double angle formulae, but more so the double sine formula, with there often being an error in that of the double cosine. However, many substituted correctly, and manipulated their expression in θ correctly using $\sin^2 \theta = 1 - \cos^2 \theta$, before converting to x . Some substituted x for $\cos \theta$ as soon as they could and sometimes got into a confused expression which they were unable to deal with convincingly. Some candidates tried to approach the problem by working on the sine and cosine parts separately; sometimes they were successful but again some became confused when they tried to put it back together in terms of x . Some candidates failed to secure the final mark by not equating to zero.

Most candidates attempted part (c) and knew that they were to factorise the cubic expression. Some used long division by the given factor of $(4x - 3)$ but then did nothing with their resulting quadratic expression, whereas others showed the correct interpretation and wrote down the factorised form of the cubic expression. Some candidates factorised the cubic expression by inspection, mostly correctly, but with the occasional error in the x -term of the quadratic factor. Some candidates who had a correct factorisation failed to score the last two marks as they simply said that the quadratic did not factorise or that it had no roots, without demonstrating why. To score the marks, candidates were required to calculate their discriminant and, if correct, note that it was negative and thus that there were no more (real) solutions to the cubic equation. Most of those who did this concluded the question with a convincing argument as to why there was only one solution to the cubic equation and thus to the trigonometric equation and stated the real solution.

Q10.

In part (a)(i), those students who used long division, perhaps misunderstanding what was meant by 'The Factor Theorem', scored no marks. Most realised the need to find the value of $p(x)$ when $x = -1$. However, it was also necessary, after showing that $p(-1) = 0$, to write a concluding statement regarding $x + 1$ being a factor; some failed to show explicitly that $-(-5) = +5$ in their verification and lost a mark; a few evaluated $p(1)$ and scored no marks.

In part (a)(ii), those who found the quadratic factor using inspection were the most successful; methods involving long division or equating coefficients usually contained algebraic errors. Once a candidate had the correct quadratic factor, it was good to see most expressing $p(x)$ as the product of three correct factors. Others found the remaining factors separately by repeated use of the Factor Theorem, but then sometimes forgot to write $p(x)$ as the product of factors. A few tried to write down the product of factors with no working at all and, although some were successful, most using this direct approach made at least one error in their factors and scored no marks at all.

In part (b), it was not acceptable to write $p(0) > p(1)$ first, simply followed by $-6 > -8$; the verification required the correct evaluation of $p(0)$ and $p(1)$ and a concluding statement such as 'therefore $p(0) > p(1)$ '. Some made errors in their calculations, usually when finding $p(1)$, but the major reason for losing marks was the omission of a suitable conclusion.

In part (c), it was pleasing to see many correct graphs, although some failed to see the connection with part (b) and showed their minimum point either to the left of or actually on the y -axis. Some caused difficulties for themselves by marking the values of the intercepts $x = -1$ and $x = 2$ equidistant from the origin; the minimum point was then almost invariably drawn in the wrong place.

Q11.

In part (a)(i), the partial fractions were found correctly by most students, substituting $x = 0$ and $x = 3$ being preferred over setting up simultaneous equations or a mix of both.

Most students went on to integrate the partial fraction form correctly, although some omitted the brackets on $\ln(x - 3)$, but this was condoned. Some students were confused by the integral of $\frac{1}{x}$, with some trying to make this become x° . Very few students produced complete nonsense for both integrals, although some differentiated.

Students attempted part (b)(i) by long division or equating coefficients and were usually successful in correctly finding all three coefficients. Those who chose equating coefficients often did far more work than was necessary, as they multiplied out and regrouped the

whole expression and were more likely to make an error. Relatively few used $x = -\frac{1}{2}$ to find r first, and those who did rarely progressed beyond a value for r .

In part (b)(ii), although most students attempted to use their result from part (b)(i), most of these failed to realise that r was not multiplied by $(2x - 1)$, and simply subtracted r from q to give a relatively simple polynomial integral, which gained no credit. Those who did use algebra correctly to set up the integral, usually gave the integral as a log term, but often with an error in the coefficient. Here too, some students differentiated rather than integrated.

Q12.

Part (a)(i) was quite often not attempted. Many candidates did not recognise the equation as that of a cubic and drew a parabola. Some drew the graph of $y = x^3$ with a stationary point of inflection but this scored no marks. The majority of candidates drew a cubic curve with a maximum point and a minimum point, but often these points were in the wrong place. Often those whose curve touched the x -axis at $x = 2$ did not draw their curve through the origin.

In (a)(ii) many candidates failed to score the mark here because of a variety of algebraic errors, often involving signs or a failure to write $= 0$ at the end of lines of working.

In (b)(i) the factor theorem was also well done but the necessary statement about $(x - 3)$ being a factor was occasionally missing. It was good to see only a few solutions using long division as they were given no credit for an incorrect method.

In part (b)(ii) the factorisation here was generally correct, and given in the required form. Many candidates were able to write down the quadratic factor by inspection.

In part (c) the more able candidates coped well with this part, establishing the value of the discriminant $(b^2 - 4ac)$ for the quadratic factor and explaining that the value of -3 was

negative and hence the quadratic equation had no real roots. Equally acceptable was an attempt to solve the equation and to comment about the negative value under the square root sign. The mark for identifying the real root as being 3 was earned by many but not by those who said that the root was $(x - 3)$ or other incorrect statements. Some weaker candidates wrongly tried to find the value of the “discriminant” using the coefficients of the cubic equation.

Q13.

Overall, this question was answered well and was a good source of marks for most candidates.

In part (a) many candidates lost a mark here through not giving a full proof of the result. Although most recognised the need to find $f(-1/3)$ many candidates merely wrote this in an unsimplified form followed by $= 0$. An extra line of working was required as well as a concluding statement.

A few candidates used the method of long division which gained no marks as the question asked for use of the Factor Theorem.

Part (b) was answered very well and most candidates obtained the correct three factors and presented them as a product. Methods used, though, varied considerably from those who just wrote the factors down by inspection, to those who used long division to find a quadratic factor and factorised that, to those who used a complete method of undetermined coefficients. All methods are valid, although some are much more efficient than others.

In (c) although there were many correct solutions, some candidates lost either a factor of 3 or x or sometimes both so could not complete correctly. Also, some stopped after correct cancelling of one factor, for which they gained some credit.

Q14.

In part (a) most candidates realised the need to find the value of $p(x)$ when $x = -1$. However, it was also necessary, after showing that $p(-1) = 0$, to write a statement that the zero value implied that $x + 1$ was a factor. It is encouraging to see an increasing number of candidates being more aware of this. Some evaluated $p(1)$ and scored no marks.

In (b)(i) those who used inspection were the most successful here; methods involving long division or equating coefficients usually contained algebraic errors.

In part (b)(ii) most candidates realised the need to consider the discriminant but several used the coefficients of the original cubic rather than those of the quadratic. Having shown that the value of the discriminant was negative it was necessary to state that the quadratic had no real roots and so the cubic had exactly one real root namely -1 .

Q15.

In part (a) virtually all candidates scored this one mark, and interpreted the result correctly for later in the question, although the interpretation was not required here. Even some of those who failed to give zero as the answer, used $(x + 2)$ as a factor.

In part (b) although most candidates knew they were to evaluate $f\left(\frac{3}{2}\right)$ with only a few attempting

$f\left(\frac{2}{3}\right)$, only about half the candidates gave a convincing response. Some assumed the evaluation would come to zero without showing any arithmetic, whilst some other candidates didn't bring their evaluation to an appropriate conclusion.

Part (c) was generally done well with correct factorisation being achieved by most candidates, although in cancelling down many then left their answer as $(2x - 1)$ rather than its reciprocal. Methods of finding the third factor in the cubic expression varied from inspection, to long division to using undetermined coefficients; the latter usually requiring far more work to achieve the result and errors often being made in attempting long division. The factorisation of the quadratic expression in the numerator was often incorrect, with many candidates who made this mistake then being unclear as to what was cancelling with what.

Q16.

This was a very straightforward question for candidates who were well versed in the necessary techniques. Even when sign errors and other small slips occurred, it was usually possible to obtain high marks. Most candidates knew that in part (b)(ii) they should mention the fact that h tends to zero, though it cannot actually be equal to zero for the question to make sense.

Q17.

In part (a), most candidates realised the need to find the value of $f(x)$ when $x = -3$. However, it was also necessary, after showing that $f(-3) = 0$, to write a statement that the zero value implied that $x + 3$ was a factor. It was good to see quite a large number of candidates being aware of this but others lost a valuable mark.

In part (b), some candidates used long division effectively to find the quadratic factor and, although this was the most successful method, some were confused by the lack of an x^2 term; others used the method of comparing coefficients or found the terms of the quadratic by inspection; a number used the Factor Theorem to find another linear factor, but seldom found both of the remaining factors.

Very able candidates were able to write down the correct product of three linear factors but many more were unsuccessful when they tried to do this without any discernible method.

Q18.

Virtually all candidates had part (a)(i) correct, with only a few making an arithmetic error. Some wrote down the implication that $(x + 1)$ was a factor of $f(x)$, which helped in part (b).

In part (a)(ii), most candidates evaluated $f\left(\frac{2}{5}\right)$ correctly, but many lost the accuracy mark either through not giving an explicit conclusion or for showing insufficient working to justify $f\left(\frac{2}{5}\right) = 0$. Some candidates chose to use algebraic division which was an acceptable method, but still required the conclusion to be drawn from a zero remainder.

In part (b), some candidates factorised by inspection using the results from part (a) and completed the question very efficiently. Others proceeded by long division and factorisation of the resulting quadratic expression to find other factors, and a few used $(ax + b)$ and determination of coefficients. Most candidates obtained the correct answer, with a great variety in the amount of working done, but a few went beyond this and were penalised for incorrect further cancelling. Some candidates thought they were required to express the answer in partial fractions.

Q19.

In part (a), most candidates knew to multiply through by the denominator and many did this correctly and went on to find A and B successfully. However, some candidates did not apparently know how to handle the 2, some omitting it altogether and some including it but not multiplying it by the denominator. Some candidates multiplied out the denominator and divided by the resulting quadratic expression to separate out the 2, then worked with the resulting algebraic fraction. Although this was unnecessary, it was a perfectly acceptable method. Relatively few candidates used the method of setting up simultaneous equations to find A and B .

In part (b), virtually all candidates knew they were to use the partial fractions to do the integration, and most recognised that log integrals were involved. The common error was

to miss the multiplier $\frac{1}{5}$ when integrating $(5x - 1)^{-1}$. Several candidates inverted the coefficient, giving it as $\frac{5}{7}$. Although most candidates integrated 2 to $2x$, many omitted the arbitrary constant.

Q20.

Part (a)(i) Those who used long division instead of the Factor Theorem scored no marks. Most candidates realised the need to show that $p(-3) = 0$. However quite a few omitted sufficient working such as $p(-3) = -27 + 63 - 21 - 15 = 0$, together with a concluding statement about $x + 3$ being a factor, and therefore failed to score full marks.

Part (a)(ii) Many candidates have become quite skilled at writing down the correct product of a linear and quadratic factor and then writing $p(x)$ as the product of three linear factors and these scored full marks. Others used long division or the Factor Theorem effectively but lost a mark for failing to write $p(x)$ as a product of linear factors. Others tried methods involving comparing coefficients, but often after several lines of working were unable to find the correct values of the coefficients because of poor algebraic manipulation. Speculative attempts to write down $p(x)$ immediately as the product of three factors were rarely successful.

Part (b)(i) This part of the question was intended to help candidates when sketching the curve. Those who found the correct values of $p(-1)$ and $p(0)$ usually scored the mark but in future a carefully written proof may be called for. Here again many arithmetic errors were seen.

Part (b)(ii) The sketch was intended to bring various parts of the question together but even very good candidates ignored the hint from part (b)(i) and showed a minimum point on the y -axis. A few lost the final mark when their curve stopped on the x -axis. Confusion

between roots and factors spoiled many sketches and several showed the y -intercept of -15 on the positive y -axis. It was disappointing that many candidates did not recognize the shape of a cubic curve at all.

Q21.

- (a) Most candidates here used the factor theorem and solved $g(x) = 0$ to find the value of d , usually successfully. Again, those candidates who used long division were more likely to make an error, particularly in interpreting the last line of working. Some candidates seemed to either not notice, or chose to ignore, the difference between $f(x)$ and $g(x)$ and used $d \pm 4 = 0$; for which they gained no credit.
- (b) Many candidates were just able to write down the required factorisation from their earlier work, particularly if it involved a correct long division,. Division here was the preferred method by those who hadn't used this technique in part (a), rather than equating coefficients; although this method was seen. Some who had made an error in part (a) often didn't notice they had an inconsistency between their value of d and their value of c .

A few candidates did part (a) backwards. They used a hybrid of methods, including trial and improvement to get a correct factorisation, realising c had to be -3 for it to work, and thus for part (a) d had to be 3 . This was quite acceptable.

Q22.

- (a) (i) The vast majority of candidates found the partial fractions correctly, most substituting the two expected values of x . Some proceeded by setting up and solving simultaneous equations and some a hybrid of both methods, but few errors were made. A few lost all the marks here through having an algebraically incorrect opening line.
- (ii) Here too, most candidates knew these were log integrals with relatively few showing no knowledge of this. The common error was to omit the ± 3 and a few candidates didn't have their log expressions in brackets and were penalised.
- (b) Various attempts at this question were seen, ranging from no attempt to a time consuming approach involving setting up simultaneous equations in P , Q and R following substitution of values of x , usually 0 , 1 and 2 . Most candidates who attempted this either set up just two equations, and thus found it was insoluble, or simply made errors in the algebra. The most successful and efficient method was algebraic long division. Relatively few candidates showed any knowledge of the technique of manipulating the numerator, as anticipated in the mark scheme. Some attempted to factorise either the numerator or denominator, or both, and made no further progress because these expressions did not factorise.

Q23.

In part (a)(i) the majority of candidates realised the need to find the value of $f(x)$ when $x = 2$. However, it was also necessary, after showing that $f(2) = 0$, to write a statement that the zero value implied that $x - 2$ was a factor. It was good to see more candidates being aware of this.

In part (a)(ii), those who used inspection were the most successful here. Methods involving long division or equating coefficients usually contained algebraic errors.

In part (b)(i) a surprising number of candidates failed to see the need to differentiate in order to find the gradient at Q . Those who attempted to find $\frac{dy}{dx}$ sometimes wrote it as $3x^2 + x$, but usually were aware of the need to substitute $x = 2$.

In part (b)(ii) those who had the correct gradient in part (b)(i) were usually successful in finding the correct equation of the tangent, and most obtained at least a method mark here.

In part (b)(iii) most were well drilled in integration and earned full marks, although some

wrote $\frac{x^4}{4} + \frac{x^2}{2} - 10$ and others gave $\frac{x^4}{4} + \frac{x^2}{2} - \frac{10^2}{2}$ as their answer. For part (b)(iv) the correct limits were usually used, although many sign/arithmetic slips occurred after substitution of the numbers 0 and 2 and it was incredible how many could not evaluate $6 - 20$ without a calculator. Very few candidates realised the need to show clearly that, although the integral from 0 to 2 gave a value of -14 , the area of the shaded region was 14. A separate statement was needed and those who simply wrote $4 + 2 - 20 = -14 = 14$ did not score full marks. Those more able candidates who made a statement about the region being entirely below the x -axis and who subsequently evaluated the integral from 2 to 0 correctly scored full marks.

Q24.

Most candidates scored full marks or close to full marks on this question, with a mark often being lost in part (a)(ii) due to lack of sufficient arithmetic being shown to justify the result or due to the conclusion from $f(x) = 0$ not being stated. Although most candidates realised parts (a) and (b) had given them two factors, relatively few could give the third factor by inspection.

Those who proceeded by a series of algebraic divisions, or called the other factor $(ax + b)$ and multiplied out and equated coefficients, paid a time penalty later.

Q25.

Part (a) was generally done very well with most candidates gaining full marks, although many methods were seen in the finding of A and B , ranging from inspection to division to rearranging the numerator. Most candidates also integrated correctly, although many omitted the constant of integration and were penalised.

In part (b), most candidates had the opening line of clearing fractions correct, although a few had an extra $(5 - x)$ and could score no further marks. The finding of P , Q and R was done accurately and concisely by many candidates. Such candidates had usually

substituted $5, -\frac{1}{3}$ and 0 for x .

The occasionally-seen error was in finding the value of Q . Those candidates who proceeded by multiplying out and equating coefficients often made numerical and/or algebraic errors and sometimes abandoned their efforts when they saw the unlikely

looking numbers emerging. This approach cost them marks and time. Some candidates used a hybrid of both methods.

All candidates who attempted part (b)(ii) knew that they were to use the partial fractions and, with the occasional exception, knew that the first two integrals were logarithmic. However, many mistakes were made in the coefficient's value and/or sign. Relatively few were successful on the third integral. Many just had the sign incorrect, although some assumed it too was a logarithm, and some expanded $(5 - x)^2$ into a quadratic expression and ended up going nowhere with it.

Q26.

Those candidates who used long division in part (a)(i) scored no marks. Most candidates realised the need to show that $p(-2) = 0$, but quite a few omitted sufficient working such as $p(-2) = -8 + 2 + 6 = 0$ together with a concluding statement about $x + 2$ being a factor and therefore failed to score full marks.

Many candidates have become quite skilled at writing down the correct product of a linear and quadratic factor and these scored full marks in part (a)(ii). Others used long division effectively but lost a mark for failing to write $p(x)$ in the required form. Others tried methods involving comparing coefficients, but often after several lines of working were unable to find the correct values of b and c because of poor algebraic manipulation.

In part (a)(iii), although many candidates tried to consider the value of the discriminant of their quadratic factor, quite a few used $a = 1$, $b = -1$ and $c = 6$ (from the cubic equation) and scored no marks for this part of the question. Others drew a correct conclusion using the quadratic equation formula, indicating that it was not possible to find the square root of -8 and others, after completing the square showed that the equation $(x - 1)^2 = -2$ has no real solutions. Some wrongly concluded that because it was not possible to factorise their quadratic then the corresponding quadratic equation had no real roots.

Most obtained the correct y -coordinate of B in part (b)(i).

In part (b)(ii) it was pleasing to see most candidates being able to integrate correctly but a large number did not answer the question set and simply found the indefinite integral in this part. Many candidates use poor techniques when finding a definite integral and it was often difficult to see the evaluation of $F(0) - F(-2)$ in their solution. Many obtained an answer of -10 which was miraculously converted into $+10$ with some comment about an area being positive. This and similar dubious working was penalized.

In part (b)(iii) some obtained an answer of -6 for the area of the triangle by using -2 as the base. Credit was given to candidates who later realised that the area of the triangle was actually 6 . Unless candidates had scored full marks in part(ii) they were not able to score full marks in this part either, even if they obtained a correct value of 4 for the shaded area.

Q27.

Many candidates were successful in this question, particularly those who used long division which was the most efficient way to solve the problem. Those candidates who chose a method based on equating coefficients often made algebraic errors. Some candidates wrote $f(x) = (ax^2 + bx + c)(3x - 1) + c$ which led to confusion and few proved the first constant c to be zero.

Q28.

In part (a)(i), a few candidates ignored the request to use the factor theorem and scored no marks for using long division. It was necessary to make a statement that “ $x + 1$ is a factor”, after showing that $f(-1) = 0$, in order to score full marks.

Part (a)(ii) was not answered as well as similar questions in previous years. Perhaps the sketch lured some into trying to write down three factors without any further working, rather than using the intermediate step of showing that $p(x) = (x + 1)(x^2 - x - 6)$ before writing $p(x)$ as a product of three factors. Many who tried long division were flummoxed by there being no x^2 term.

In part (b)(i), those who had the correct linear factors in part (a)(ii) usually wrote down correctly that A had coordinates $(-2, 0)$, although some carelessly wrote the point as $(0, -2)$.

Many candidates simply found an indefinite integral in part (b)(ii) and then a definite integral in part (b)(iii). The two parts were generously treated holistically when candidates did this. The fractions once again caused problems to most candidates who are so used to having a calculator to do this work for them. It was very rare to see the correct answer of -32 for the definite integral.

In part (b)(iii), many lost out on an easy mark because they rolled their two sections into one: those who wrote “integral = $-32 = 32$ ” gained full credit for part (b)(ii) but did not score the mark in part (b)(iii). It was necessary to give a positive value for the area of the region and to make this explicit. In anticipation of a lot of wrong answers in part (b)(ii), a follow through mark was awarded in part (b)(iii): for example, if a candidate’s answer in part (b)(ii) was -20 and they concluded that the area was 20 in part (b)(iii), they scored the mark.

In part (b)(iv), most candidates differentiated correctly, but quite a few thought that $3(-1)^2 - 7$ was equal to -10 and thus obtained the wrong gradient of the curve.

In part (b)(v), a large number of candidates found the correct equation of the normal but some still confused tangents and normals and consequently thought that the gradient of the normal was equal to -4 . It was quite common for weaker candidates to either negate their gradient or take the reciprocal but to fail to do both.

Q29.

Most candidates scored well on this question although there were some unnecessarily long solutions.

In part (a)(i), most candidates substituted $x = \frac{1}{2}$, correctly into the polynomial, but many lost a mark by not demonstrating that the result came to zero by showing the required arithmetic and/or by not interpreting the result as demonstrating that $(2x - 1)$ is a factor. For part (a)(ii), many candidates did an algebraic long division or multiplied out the given expression and equated the coefficients. Both techniques lead to the correct answer, but the value of q could be written down by inspection and the value of p should follow in a couple of lines of working. This is where many candidates spent an unnecessarily long time in their working. Most candidates answered part (a)(iii) correctly, although, surprisingly, the common error was to just omit the 4 when factorising the numerator.

Part (b) was attempted via a variety of methods. The most concise of these was to divide $2x^2$ by $x^2 + 2x - 15$ and interpret the remainder. Many candidates who took the more conventional partial fractions approach had not realised that $A = 2$ was immediate and so they set up simultaneous equations in three unknowns, often making a mistake in their solution.

Those who substituted suitable values of x were usually more successful. Some

candidates wrote the right-hand side as $\frac{B}{x+5} + \frac{C}{x-3}$ with some then going on to find the values of B and C correctly and recombining their fractions to get the required form. However, this is an unnecessarily long route to the solution, and mistakes were often made along the way, not least in the opening line when candidates were attempting to multiply through by the denominator.

Q30.

Many candidates lost a mark in part (a)(i) for failing to make a statement regarding the factor: they assumed that it was sufficient just to show that $p(-2) = 0$.

In part (a)(ii), those who found the quadratic factor $x^2 - x - 6$ by long division usually went on to earn full marks. Some used the Factor Theorem but the repeated factor was problematic. Those who used inspection or comparing coefficients generally scored at least one mark. Not all candidates wrote the final answer as the product of three linear factors and were possibly confused by the repeated factor.

In part (b)(i), almost all candidates stated the value of k as -12 , but a few wrote $+12$.

Most candidates found the graph in part (b)(ii) to be a difficult one to draw. Parabolas through $(-2, 0)$ and $(3, 0)$ were common and scored no marks. Those who drew a cubic shape usually had three intercepts on the x -axis and this was awarded one mark. Once again, the repeated root confused many. Some of those who made a good attempt lost a mark for having the minimum point on the y -axis.

Q31.

In part (a) most candidates showed sufficient arithmetic to justify the given result; those

who simply wrote down $f\left(-\frac{2}{3}\right)$ without any evaluation seen were not awarded this mark.

In part (b) most candidates did achieve the correct answer to this question, although it often was not clear how they had got to it. The clearest and most efficient method was used by those who divided $f(x)$ by $(2x + 3)$ and factorised the resulting quadratic expression. Those candidates who used a method based on determining coefficients often assumed a quadratic expression $ax^2 + bx + c$ and took time to fully multiply this by $2x + 3$ when the values of a and c should have been immediate. Others proceeded by, essentially, trial and improvement, often dividing by $3x + 1$, apparently not noting from part (a) that this could not be a factor.

For part (c) full marks were dependent on candidates being able to factorise the quadratic denominator, which many did successfully, although many also made an error. Those who got to the simplified result usually displayed it clearly, whereas others left an algebraic

fraction apparently abandoned.

Q32.

Part (a)(i) Most candidates found $p(-2)$ but often failed to convince examiners that they had really shown that $k = 10$. Many substituted $k = 10$ from the outset and then drew no conclusion from the fact that $p(-2) = 0$. Those using long division often made sign errors.

Part (a)(ii) Factorisation of a cubic seems well understood and, apart from those who could not factorise $x^2 - 6x + 5$, candidates usually scored full marks. Some still confuse factors and roots.

Part (b) The sketch was generously marked with regard to the position of the stationary points but it was expected that candidates would indicate the values where the curve crossed the coordinate axes and often these values, particularly the 10 on the y -axis, were omitted.

Q33.

This question was done well with again many candidates gaining full marks. In part (a), those who explicitly used the Factor Theorem with $g(x)$ gave the clearest explanations. Those who tried to explain the given result without using the Factor Theorem found themselves resorting to the word “must”, usually ending with d must be -4 , without giving a convincing explanation as to why.

There was also some confusion in notation between f and g for the functions involved. It was anticipated that part (b) would be done by inspection, but the preferred method was algebraic division, usually done accurately. Some candidates multiplied out and equated coefficients, usually correctly, but this method does involve some unnecessary work; $b = -3$ should be immediate.

Q34.

This question was generally done well with many candidates gaining full marks. Marks tended to be lost in part (a) rather than part (b), possibly because the latter was a more conventional partial fractions problem. There were several different approaches to part (a) including manipulating the numerator to contain $A(x - 3)$, algebraic division and cross multiplying and using partial fraction type techniques, or some mix of these. Errors tended to be algebraic or numerical. In part (a)(ii) it was notable that some candidates did not attempt a logarithmic integral, although they went on to do so in part (b).

In part (b) virtually all candidates scored well, either using the expected two values of x or equating coefficients to find the values of P and Q . Again a few made algebraic or numerical errors, with very few cross multiplying incorrectly. Similarly virtually all candidates went on to give some form of logarithmic integrals, sometimes with errors in the coefficients. Very few non-sensible attempts at the integral were seen.

Q35.

Part (a)(i) Most candidates realised the need to find the value of $f(x)$ when $x = 1$. However, it was also necessary, after showing that $f(1) = 0$, to write a statement that the zero value implied that $x - 1$ was a factor.

Part (a)(ii) Those who used inspection were the most successful here. Methods involving long division or equating coefficients usually contained algebraic errors.

Part (a)(iii) This section seemed unclear to some candidates. Many tried to find the discriminant but used the coefficients of the cubic equation. Many who used the quadratic thought that in order to have one real root the discriminant had to be zero, no doubt thinking the question was asking about equal roots. Some correctly stated that 1 was the only real root but many were obviously confused by the terms “factor” and “root” and stated that “ $x - 1$ was a root”.

Part (b)(i) Most candidates were well versed in integration and earned full marks here.

Part (b)(ii) The correct limits were usually used, although many sign/arithmetic slips occurred after substitution of the numbers 1 and 2. Very few candidates realised the need to find the area of a triangle as well and so failed to subtract the value of the integral from the area of the triangle in order to find the area of the shaded region.



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